Actividad 4.1

21/04/2026

$x = 10 \text{ a } 51$

Trayectoria 1. $f(x) = 2\,\text{sen}(x^2)$

① $x(t) = t$

$y(t) = 2\,\text{sen}(x^2)$

② $\dot{x}(t) = 1$

$\dot{y}(t) = 4x\cos(x^2)$

③ $u(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$

$u(t) = \sqrt{1 + 4x\cos(x^2)^2}$

④ $\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right)$

$\theta(t) = \tan^{-1}(4x\cos(x^2))$

⑤ $w(t) = \left(\frac{d\theta}{dt}\right)$

$w(t) = \left(\frac{d(\tan^{-1}(4x\cos(x^2))}{dt}\right)$

$d(4x\cos(x^2)) = 4(\cos(x^2) - 2x^2\sin(x^2))$

$= 4\cos(x^2) - 8x^2\sin(x^2)$

$w = \frac{4\cos(x^2) - 8x^2\sin(x^2)}{1 + 4x\cos(x^2)^2}$

para condiciones iniciales:

se sustituye 0 en las originales

$x(0) = t$

$= \emptyset$

$y(0) = 2\,\text{sen}(0)$

$= \emptyset$

$\theta(\emptyset) = \tan^{-1}(4(0)\cos(0)$

$\theta(\emptyset) = \tan^{-1}(4(0)(1)$

$\theta(\emptyset) = \tan^{-1}(0)$

$\theta(\emptyset) = \emptyset$

$x = |-4 \text{ a } 4|$

trayectoria 2 $\quad f(x) = x^2 + y^2 - 16$

1. $x(t) = t$

   $y(x) = \sqrt{16 - t^2}$

2. $\dot{x}(t) = 1$

   $\dot{y}(t) = \dfrac{-x}{\sqrt{16 - x^2}}$

3. $u(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$

   $u(t) = \sqrt{1 + \left(\dfrac{-x}{\sqrt{16 - x^2}}\right)^2}$

4. $\theta(t) = \tan^{-1}\left(\dfrac{\dot{y}}{\dot{x}}\right)$

   $\theta(t) = \tan^{-1}\left(\dfrac{-x}{\sqrt{16 - x^2}}\right)$

5. $w(t) = \left(\dfrac{d(\theta)}{dt}\right)$

   $w(t) = \left(\dfrac{d\left(\tan^{-1}\left(\frac{-x}{\sqrt{16 - x^2}}\right)\right)}{dt}\right)$

   $d\left(\dfrac{-x}{\sqrt{16 - x^2}}\right) = \dfrac{-16}{(16 - x^2)(\sqrt{16 - x^2})}$

condiciones iniciales

se sustituye 0 en las originales

$x(-4) = t$

$= -4$

$y(-4) = \sqrt{16 - (-4)^2}$

$$w = \frac{\dfrac{-16}{(16 - x^2)(\sqrt{16 - x^2})}}{\sqrt{1 + \left(\dfrac{-x}{\sqrt{16 - x^2}}\right)^2}}$$